n8n Exercise: RSS Feed → AI Summary → Google Sheets

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Overview

By the end of this exercise you will have a working n8n workflow that reads an RSS feed, summarizes each article with a free AI model from OpenRouter, and writes the results into a Google Sheet.

The workflow has four steps:

1. **Manual Trigger** starts the run when you click a button.
2. **RSS Read** pulls the latest articles from a feed.
3. **Basic LLM Chain** sends each article to an OpenRouter model and gets back a 2-sentence summary.
4. **Google Sheets** appends one row per article: title, link, date, summary.

Before you start, you need:

- n8n running locally and open at `http://localhost:5678`
- A free OpenRouter account and API key
- A Google account (for Sheets and Google Cloud)

Part 1: Prepare the Google Sheet

Create the sheet first so n8n has a destination with the right column names.

1. Go to sheets.google.com and create a blank spreadsheet.
2. Rename it to `RSS Summaries`.
3. In row 1, type these four headers, one per column:

```
4.
```

Part 2: Build the workflow

You will add the nodes one at a time and test each one before moving on.

Step 1: Create the workflow and trigger

1. In n8n, click **Create workflow** and name it `RSS Summarizer`.
2. Click **Add first step** and choose **Manual Trigger**. It appears as "When clicking 'Execute workflow'".

Step 2: Read the RSS feed

1. Click the **+** after the trigger and search for **RSS Read**.
2. Set **URL** to `https://feeds.bbci.co.uk/news/technology/rss.xml`. Any other RSS feed works too.

3. Click the Execute **step**. The output panel should list articles with `title`, `link`, `pubDate` and `contentSnippet`.

Step 3: Limit the number of items

1. Click **+** after RSS Read and add a **Limit** node.
2. Set **Max Items** to `5`.

This keeps test runs fast and stays within free-model rate limits.

Step 4: Add the Basic LLM Chain

1. Click **+** after Limit and add **Basic LLM Chain** (under AI).
2. Set **Source for Prompt** to *Define below*.
3. In the **Prompt** field, switch to **Expression** and paste:

```
Summarize this article in 2 plain-English sentences.

Title: {{ $json.title }}
Content: {{ $json.contentSnippet }}
```

The node will show a missing-model warning. You fix that in Part 3.

Part 3: Connect a free OpenRouter model

The summaries come from a free model on OpenRouter, so no payment is needed. Free model IDs end in `:free`, and OpenRouter also offers `openrouter/free`, a router that picks one available free model at random (OpenRouter docs).

1. Create an API key at openrouter.ai/keys and copy it.
2. On the Basic LLM Chain node, click the **+** under **Model** and choose **OpenRouter Chat Model**.
3. Under **Credential**, click **Create new credential**, paste the API key, and save.
4. Set **Model** to one of these:
    - `openrouter/free` for the simplest setup. Summary style may vary from article to article.
    - A specific model ID ending in `:free`, copied from the free models list. Pick one for consistent output.
5. If the name is not in the dropdown, switch the field to **Expression** and type the ID.
6. Click **Execute step** on the Basic LLM Chain. Each item should now have a `text` field with the summary.

Free models also have lower rate limits and can be slower at peak times.

Part 4: Connect Google Sheets

n8n running locally needs its own Google OAuth credentials. The one-time setup below takes about 10 minutes.

Step 1: Create the Google credentials

1. Open Google Cloud Console and create a new project.

2. Go to **APIs & Services → Library** and enable both the **Google Sheets API** and the **Google Drive API**.
3. Go to **APIs & Services → OAuth consent screen**, choose **External**, fill in the required fields, and add your own Google account under **Test users**.
4. Go to **APIs & Services → Credentials → Create credentials → OAuth client ID** and choose **Web application**.
5. Under **Authorized redirect URIs**, add exactly: `http://localhost:5678/rest/oauth2-credential/callback`
6. Click **Create** and copy the **Client ID** and **Client Secret**.

Step 2: Add the credential in n8n

1. Back in the workflow, click **+** after the Basic LLM Chain and add a **Google Sheets** node.
2. Under **Credential**, click **Create new credential** and choose **Google Sheets OAuth2 API**.
3. Paste the **Client ID** and **Client Secret**, then click **Sign in with Google**.
4. Approve access. If Google warns the app is unverified, continue; this is expected for your own test project.

Step 3: Configure the node

1. Set **Resource** to *Sheet Within Document* and **Operation** to *Append Row*.
2. **Document:** choose *From list* and select `RSS Summaries`.
3. **Sheet:** select `Sheet1`.
4. Set **Mapping Column Mode** to *Map Each Column Manually*.
5. Fill each column with an expression:

Column	Expression
Title	`{{ $('RSS Read').item.json.title }}`
Link	`{{ $('RSS Read').item.json.link }}`
Published	`{{ $('RSS Read').item.json.pubDate }}`
Summary	`{{ $json.text }}`

The Basic LLM Chain only passes along `text`, which is why Title, Link and Published are pulled back from the RSS Read node by name.

Part 5: Execute and verify

A successful run shows a green tick on every node and 5 new rows in your sheet.

1. Click **Execute workflow** at the bottom of the canvas.
2. Watch the nodes run left to right. Each should show a green tick and an item count (5 items through RSS Read, Limit, the chain and Google Sheets).
3. Click the Google Sheets node and confirm its output lists 5 rows.
4. Open the `RSS Summaries` sheet. Each row should have a title, link, date and a 2-sentence summary.

5. Press **Ctrl/Cmd + S** to save the workflow.

Handling rate limits in a live class

Free models can reject requests when many people run the workflow at once, so build in retries before the session.

1. Open the Basic LLM Chain node, go to **Settings**, and turn on **Retry On Fail**. Try 3 tries with a 2000 ms wait.
2. Keep the **Limit** node at 5 items or fewer.
3. Optionally turn on **Continue On Fail** in the same settings so one failed summary does not stop the whole run.
4. If many students hit errors together, have them run the workflow one at a time or stagger their starts.

Optional: run it on a schedule

To run the workflow automatically, replace the Manual Trigger with a **Schedule Trigger** (for example, daily at 8:00) and switch the **Active** toggle on.

Each scheduled run re-appends articles that are still in the feed. To avoid duplicates, change the Google Sheets operation to *Append or Update Row* and match on the `Link` column.

Troubleshooting

Problem	Likely cause and fix
403 or "credential does not have access" from Sheets	The Sheets or Drive API is not enabled, or your Google account is not listed under Test users.
Redirect URI mismatch when signing in	The URI in Google Cloud must match exactly, including `http` and the port: `http://localhost:5678/rest/oauth2-credential/callback`.
Summary column is empty	The Summary field must be `{{ $json.text }}`, not an RSS field name.
OpenRouter 401 error	The API key is wrong or was pasted with extra spaces. Create the credential again.
OpenRouter 429 or model not found	The free model is rate-limited or has left the free catalog. Retry, or choose another `:free` model or `openrouter/free`.
Model name missing from the dropdown	Switch the Model field to **Expression** and type the ID manually.